MT1011 ENGINEERING MATERIALS

MIDTERM REFERENCE • 2025 PAPER-READY EDITION

Name: ..... Student ID: .....

01 ATOM / ELECTRONS

Atomic number Z = proton count; neutral atom electrons = Z. **Mass number** ≈ protons + neutrons (isotope); **atomic weight A** is average molar mass, NOT the integer mass number. Neutrons = mass number − Z. A cation has fewer electrons, an anion more.

Orbital filling: 1s → 2s → 2p → 3s → 3p → 4s → 3d. Capacities s^2, p^6, d^{10}. When transition metals ionise, remove 4s electrons **before** 3d. Fe: [Ar] $4s^2 3d^6$; Fe^{2+}: [Ar] $3d^6$. Ni: [Ar] $4s^2 3d^8$; Ni^{2+}: [Ar] $3d^8$.

Material families: metals (ductile, conductive), ceramics (hard/stiff, brittle, heat-resistant), polymers (chains, low density), composites (matrix + reinforcement); structure/processing govern properties.

02 BONDS / IONICITY

Ionic: electron transfer + opposite ion electrostatic attraction, nondirectional; **covalent:** shared valence electrons, directional; **metallic:** positively charged cores + delocalised electrons; **secondary:** van der Waals, dipoles, hydrogen (typically weaker). Ceramics: mainly ionic and/or covalent.

$$\%IC = [1 - e^{-0.25(\chi A - \chi B)^2}] \times 100\%$$

χA, χB = Pauling electronegativities; use absolute Δχ; %IC ionic character. Greater Δχ → more ionic; smallest Δχ → most covalent. For Cu compounds compare Cu χ=1.9 with F 4.0; Cl 3.0; S 2.5; O 3.5.

$$F_N = F_A + F_R \; ; \; F_N(r_0) = 0$$

F_A, F_R signed attraction/repulsion; r_0 equilibrium spacing. Energy minimum at r_0; deep energy well → strong bonding; curvature → stiffness; asymmetry → expansion.

WATCH: Hydrogen bonding is secondary and weaker than primary metallic, ionic, covalent bonding (individual-bond comparison).

03 CRYSTALS / CELL DATA

Crystalline: long-range ordered lattice; **amorphous**: lacking long-range order. Unit cell repeats in 3D. Lattice parameters a,b,c (edges), α,β,γ (angles). 7 crystal systems; 14 Bravais lattices. **SC/BCC/FCC** are cubic; **HCP** hexagonal close packed.

Cell contributions: corner ⅛, face ½, edge ¼, body 1. **SC:** n1, CN6, APF0.52, a=2R. **BCC:** n2, CN8, APF0.68, √3a=4R. **FCC:** n4, CN12, APF0.74, √2a=4R. **HCP:** n6 (conventional), CN12, APF0.74, ideal c/a=1.633.

$$\rho = nM / (N_A V_c) \; ; \; APF = n(4\pi R^3/3)/V_c$$

ρ density; n atoms/cell; M molar mass; N_A=6.022×10²³ mol^{-1}; V_c=a^3 (cubic); R atomic radius. Keep units consistent.

03 CRYSTALS / CELL DATA (CONT.)

Cubic directions/planes: FCC close-packed {111}⟨110⟩; BCC dense ⟨111⟩. Slip easiest on dense plane/direction. ⟨uvw⟩ family directions; {hkl} family planes; (hkl) plane; [uvw] direction.

04 INDICES / COORDINATES

Direction [uvw]: (1) arrow head − tail in **fractional** x,y,z cell coordinates; (2) clear fractions with common factor; (3) reduce to smallest integers; (4) negatives use overbar. Translate the vector; index describes orientation, not location.

Plane (hkl): (1) intercepts with axes in a,b,c units; (2) if through origin translate parallel plane; (3) reciprocals of intercepts; (4) clear fractions; (5) use overbars for negatives. Parallel to axis → intercept ∞ → index 0.

$$hx + ky + lz = 1 \text{ (fractional coordinates)}$$

For a cubic plane (hkl) in intercept form with suitable origin; x,y,z are coordinates divided by lattice constants. For (110), x+y=1; z arbitrary. Test point by substitution.

$$d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$$

d_{hkl} plane spacing in cubic lattice; a cell edge; h,k,l plane indices. {111} families differ only in signs/permutations of 1,1,1.

WATCH: Do not mistake fractional coordinates (½,½,1) for direction [112]. Miller plane and direction brackets mean different objects.

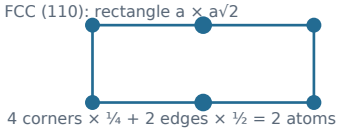
05 LINEAR / PLANAR DENSITY

$$LD = N_{\text{on line}} / L \; ; \; PD = N_{\text{on plane}} / A$$

LD atoms per nm; PD atoms per nm². Count atom **centres** actually on the segment/plane and fractional sharing at its boundary.

BCC [111]: segment body diagonal L=√3a=4R; count ½ +1 +½=2 atoms; LD = 2/(4R) = 1/(2R).

FCC (110): planar rectangle a × a√2; effective atoms 4×¼ + 2×½ = 2; PD = 2/(√2a^2) = √2/a^2 = 1/(4√2 R^2).



Useful FCC: PD(100)=2/a^2; PD(111)=4/(√3a^2); LD[100]=1/a, LD[110]=2/(√2a), LD[111]=1/(√3a). **Cubic** use a↔R contact rule for its own structure.

06 DEFECTS / SLIP

06 DEFECTS / SLIP (CONT.)

Point: vacancy, self-interstitial, substitutional (foreign atom at regular lattice site), interstitial (small atom between sites). **Line:** edge dislocation (extra half plane, b⊥line), screw (helical shear, b∥line), mixed. **Planar:** grain boundaries, twins, stacking faults, free surfaces.

Substitutional identification: different atom occupying a normal lattice point, even if strain distorts neighbors. Vacancy = empty lattice site; interstitial = added atom away from usual sites. **Plastic slip:** dislocation motion, not vacancy motion.

$$N_v = N \exp[-Q_v / (k_B T)]$$

N_v equilibrium vacancies; N atomic sites; Q_v vacancy formation energy; k_B=8.617×10⁻⁵ eV/K or 1.381×10⁻²³ J/K; T kelvin.

Burgers vector: closure failure of atom-step circuit around a dislocation. Slip system = slip plane plus slip direction. FCC has 12 {111}⟨110⟩ slip systems.

07 MECHANICAL TESTS

Tensile: engineering σ=F/A_0; engineering ε=(L−L_0)/L_0. Original A_0 and L_0, load F, current gauge length L. Linear elastic σ=Eε, E Young modulus. Yield σy = onset of plastic strain; 0.2% proof offset = 0.002 strain. UTS=Fmax/A_0; necking follows UTS; fracture afterwards.

$$\sigma_{true} = F/A_{inst} \; ; \; \varepsilon_{true} = \ln(1+\varepsilon)$$

Before necking, uniform constant-volume plastic flow: $\sigma_{true} \approx \sigma(1+\varepsilon)$. A_{inst} instantaneous area. After necking, simple conversion fails.

$$\%EL = 100(L_f - L_0)/L_0 \; ; \; \%RA = 100(A_0 - A_f)/A_0$$

L_f final gauge length; A_f final neck area. Measures ductility; sample length affects %EL.

$$U_r = \sigma_y^2/(2E) = \tfrac{1}{2}\sigma_y \varepsilon_y$$

U_r modulus of resilience = elastic area under stress–strain. Toughness = entire area to fracture; stiffness E ≠ strength σy ≠ toughness. Rockwell depth; Vickers pyramid; Brinell ball; Charpy notched impact energy.

08 STRENGTHENING

Core rule: hinder dislocation motion → greater strength/hardness, often less ductility. (1) Grain refinement increases grain boundary obstacles. (2) Cold work raises dislocation density. (3) Solid-solution solute strain fields impede slip. (4) Fine precipitates obstruct slip.

$$\sigma_y = \sigma_0 + k_y/\sqrt{d} \text{ (Hall–Petch)}$$

σy yield strength; σ_0 friction stress; ky material coefficient; d average grain diameter. Smaller d → higher σy (usual valid grain scale).

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08 STRENGTHENING (CONT.)

$$\%CW = 100(A_0 - A_d)/A_0$$

A_0 area before deformation; A_d area after; greater cold work → hardness/strength ↑, ductility ↓. Recovery → recrystallization → grain growth on annealing.

WATCH: PURE metal: grain refinement and work hardening possible; solid-solution strengthening requires solute (not chemically pure). Reducing grain-boundary area generally weakens grain-boundary strengthening. 2025 Q23 has ambiguous distractor A and professor-keyed D.

09 SOLID SOLUTION / PHASES

Solute/solvent: minority atoms / host. Hume–Rothery extensive substitutional miscibility: atomic radius difference ~<15%, same structure, similar electronegativity, same valence. **Phase:** homogeneous composition/structure; **solubility limit:** largest amount dissolved at a temperature. **Microstructure:** shapes, sizes and amounts of phases/grains.

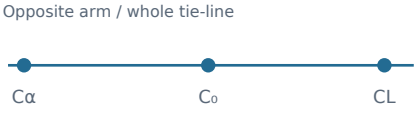
Phase diagram: equilibrium stable phases vs T and overall composition C_0. **Liquidus** boundary first solid on cooling; **solidus** last liquid freezes; **solvus** solid solubility boundary. **Cu–Ni** complete FCC solid solubility (L / α+L / α).

10 TIE LINE / LEVER RULE

Method: ① mark overall C_0 & T; ② region and phase names; ③ horizontal tie line; ④ read end-point compositions C_{left}, C_{right}; ⑤ opposite-arm fractions; ⑥ fractions sum 1; ⑦ compositions are tie-line ends, NOT overall C_0.

$$W_{left} = (C_{right} - C_0)/(C_{right} - C_{left})$$

$W_{right} = (C_0 - C_{left})/(C_{right} - C_{left})$; phase mass fractions. C in wt% of element named on bottom axis.



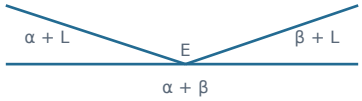
Always use **same composition units** (wt% vs at%). If right solid = pure Ge, C_{solid} ≈ 100 wt.% Ge. For L + Ge: $W_L = (100 - C_0)/(100 - C_L)$.

11 EUTECTIC SYSTEMS

$$L(C_E) \rightarrow \alpha(C_{\alpha E}) + \beta(C_{\beta E})$$

At T_E on cooling; C_E eutectic liquid composition; α/β solubility-endpoint solids. Hypoeutectic $C_0 < C_E$: primary α + eutectic α+β; hypereutectic $C_0 > C_E$: primary β + eutectic α+β.

11 EUTECTIC SYSTEMS (CONT.)



Cu–Ag (wt% Ag): E=71.9 wt% Ag, 779°C; $C_{\alpha E}$=8.0, $C_{\beta E}$=91.2. α Cu-rich FCC; β Ag-rich FCC. **Pb–Sn (wt% Sn):** E=61.9%, 183°C; $C_{\alpha E}$=18.3, $C_{\beta E}$=97.8; pure Pb 327°C, Sn 232°C.

Cooling microstructure: pure L → L+primary solid on first liquidus crossing → final remaining liquid becomes α+β eutectic at T_E. Eutectic microconstituent fraction = liquid fraction immediately above T_E, **not** total α or β phase fraction.

12 Au-Ge 2025 PAST PAPER

Figure 4, horizontal = wt% Ge: melting Au 1064.43°C; Ge 938.3°C. Eutectic 12.5 wt.% Ge at **361°C**; near Au terminal solubility ≤1 wt.% Ge; Ge-rich solid ≈ pure Ge. Diagram must be read with its **own** endpoints, not Cu–Ag/Pb–Sn numbers.

75 wt.% Ge cooling: on Ge-rich side, primary Ge first when vertical composition reaches Ge-side liquidus (≈900°C); all remaining liquid eutectic-freezes at 361°C; final microstructure primary Ge + Au–Ge eutectic. C_L at temperature = liquidus intersection of horizontal tie line (≈32.5 wt% Ge at 700°C).

Fraction at 800°C: Ge-rich primary solid C_s≈100 wt% Ge and C_L read at 800°C (~46 wt% Ge); apply $W_L = (C_s - C_0)/(C_s - C_L)$. Read graph, do NOT interpolate using atomic-% top scale.

13 EXAM ROUTING / TRAPS

Electronic configuration: remove 4s first for metal cations. **Bond % ionic:** electronegativity → square Δχ → exponent → ×100. **Covalent ranking:** smallest Δχ. **Defects:** substitutional = on lattice site; plastic slip = dislocations.

Diagram: indices [uvw] use head−tail; (hkl) use intercept reciprocals; {111} all ±1. For (110) membership, x+y=1 in fractional cell coordinates. **Atomic density:** count on line/plane, do not use n per unit cell directly.

Phase diagram: read axes (wt% vs at%), locate C_0, cross first liquidus, eutectic horizontal at final freeze, tie line to get phase composition, opposite-arm fraction; post-freeze microconstituents ≠ phase fraction. Verify dimensions and probability-like fractions 0–1.

WATCH: 2025 professor's marked solutions include Q23 = D, but option A also does not increase strength. Interpret it as "method unavailable in chemically pure metal" to select D.

14 7 SYSTEMS / 14 BRAVAIS

Cubic: a=b=c; α=β=γ=90°; P, I, F (3 lattices). **Tetragonal:** a=b≠c, all angles 90°; P, I (2). **Orthorhombic:** a≠b≠c, all angles 90°; P, C, I, F (4). **Hexagonal:** a=b≠c, α=β=90°, γ=120°; P (1).

Trigonal / rhombohedral: a=b=c, α=β=γ≠90°; R (1). **Monoclinic:** a≠b≠c, α=γ=90°, β≠90°; P, C (2). **Triclinic:** a≠b≠c, generally unequal non-right angles; P (1). TOTAL = 14 Bravais lattices.

P primitive (lattice points at corners); **I** body-centred; **F** all-face-centred; **C** base-centred; **R** rhombohedral. A Bravais lattice describes periodicity; a **basis** of atoms attached to each point defines actual crystal structure.

15 SLIP / ATOMIC GEOMETRY

SC: contact along edge; BCC along body diagonal; FCC along face diagonal. HCP and FCC are close packed, with stacking sequences ABAB and ABCABC respectively; 12 nearest neighbours. Pure Fe below ~910°C is BCC; above ~910°C it is FCC (up to the next transformation).

Structure preference: FCC usually high room-temperature ductility because {111} close-packed planes have multiple active directions. BCC has no strictly close-packed plane and its slip is more temperature sensitive. HCP can have fewer readily activated slip systems at low temperature.

$$\tau_R = \sigma \cos\varphi \cos\lambda$$

Schmid relation for resolved shear on a slip system: σ axial applied stress; φ angle between stress axis and slip-plane normal; λ angle between stress axis and slip direction. Slip when τ_R reaches CRSS.

Cubic direction length: $L = a\sqrt{(u^2+v^2+w^2)}$ for an integer lattice translation [uvw]. **Planar packing:** highest-density FCC plane {111}; BCC dense direction <111>. **Density unit check:** atoms/nm vs atoms/nm² vs g/cm³.

16 PROPERTY DISTINCTIONS

Stiffness E = elastic stress/strain slope; **yield σy** = plastic onset; **UTS** = maximum engineering stress; **ductility** = plastic elongation; **toughness** = entire stress–strain area; **resilience** = elastic area; **hardness** = indentation resistance. These properties are NOT interchangeable.